# Dapeng WANG (Phoenix)

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## EDUCATION

**THE HONG KONG POLYTECHNIC UNIVERSITY** — Hong Kong, China

*Bachelor of Engineering (Honors) in Mechanical Engineering* — September 2020 – July 2024

- GPA: 3.80/4.3 (3.70/4.0 from WES calculation) (First Class Honour)
- Major GPA: 3.87/4.3
- Dean's Honours List 2021/22, 2022/23 (Top 10%)
- The Hong Kong Polytechnic University Scholarship 2023/24

## RESEARCH EXPERIENCE

**ROBOTIC FISH (CAPSTONE PROJECT)** — Hong Kong, China

September 01, 2023 - May 13, 2024

- Design and develop a robotics fish under the supervision of Dr. Hui Tang
- The fish could successfully thrusting, diving & lifting.
- My jobs are dynamics modelling of lifting & diving and fish tail, building Kalman Filter for velocity and altitude control, signal processing and hardware design & manufacturing.
- This project received A+ as final grade (10% A+ rate).

**STUDENT RESEARCH ASSISTANT IN RENEWABLE ENERGY ADVANCEMENT LAB** — Hong Kong, China

*Research Assistant* — October 03, 2023 - January 31, 2024

- Conduct research on the continued work of PIML-DCNN model on quantify staggered-height configurations on convective heat transfer of PV arrays from URIS under the supervision of Dr. Mengying LI.
- Paper under review, 'Fast estimation of the convective cooling performance of Photovoltaics array with various configurations: a physics informed machine learning approach' to 'Enery & AI' as the first author.

- Helping the Lab apply for applying hyper computation platform of The Hong Kong Polytechnic University.
- Helping the Lab daily affairs, like help writing grants and reports.

**UNDERGRADUATE RESEARCH AND INNOVATION SCHEME (URIS)** — Hong Kong, China

*Undergraduate researcher in scheme* — June 07, 2022 – August 31, 2023

- Conduct research on the convective heat transfer effect of staggered-height arrangement on PV farms under the supervision of Dr. Mengying LI.
- Submitted a conference paper, "Efficiency improvement of solar Photovoltaic arrays using height-staggered configurations", to PolyU Research Student Conference (PRSC 2023), and the abstract has been accepted.
- Proposed a new method with Physics-Informed Machine Learning and Deep Convolution Neural Network combined model (PIML-DCNN) to solve the interpretability of PV arrays staggered-height configurations on convective heat transfer in quantitative view. Such method and relative experiment validation results has been recorded in URIS Final Report.
- The final report of URIS was rated 'very satisfactory' by the committee.

**STUDENT RESEARCH ASSISTANT IN RENEWABLE ENERGY ADVANCEMENT LAB** — Hong Kong, China

*Research Assistant* — Jan 24 - April 29, 2022

- Conduct research on adaptive cooling method combining radiative and convective heat transfer of photovoltaic panels under the supervision of Dr.LI Mengying.
- Building prototypes for flapping convective heat transfer of PVs and conducting experiment.

**2021 SPRING CETUS-TALK ONLINE RESEARCH** — Hong Kong, China

June 06 – Augst 26, 2021

- Conduct research about a typical autonomous driving navigation algorithm under the supervision of Prof. Fumiya Lida from Cambridge University and get the recommendation letter from him.
- Published the conference paper, "New Improvement of 'Dynamic Repulsive Potential Factor' on Artificial Potential Field Path Planning Algorithm" on 2021 2nd IYSF Academic Symposium on Artificial Intelligent and Computer Engineering.

## EXTRA-CURRICULAR ACTIVITIES

**MATHEMATICS FOR ENGINEERING ONLINE SUMMER PROGRAMME (ONLINE MODE)**

Hong Kong, China

July 26 – July 30, 2021

- Study linear algebra, numerical analysis from Dr. Nick Godfrey from Girton College, Cambridge.

**JOINING HONG KONG REGIONAL COMPETITION 2022 OF ENACTUS** Hong Kong, China
*13th President of Enactus HKPU & Project Manager* May 24, 2021 – June 25, 2022

- Working as 13th president of Enactus HKPU and project manager of team 'Snakecup', preparing for the regional competition of year 2022.
- Developing a template app service for our project by studying Javascript and html by myself.
- Distributing workload of the project based on teammates' studying major and personalities.
- Successfully joining and finishing 2022 regional competition of Enactus HK.

**2022 GLOBEX JULMESTER (ONLINE MODE)** Beijing, China
*Exchange student* July 04 – July 22, 2022

- Attending 2022 Globex Julmester held by Peking University.
- Studying Intelligent Manufacturing and Service Systems from Professor Andrew KUSIAK, and Control Theory from Professor. Xun HUANG.
- Achieve high scores of classes: Intelligent Manufacturing and Service Systems: 97, Control Theory: 92.

**PROOF-OF-CONCEPT (POC) FUNDING SCHEME 2022** Hong Kong, China
*Start-up team member, CTO* October 19 – November 28, 2022

- Working as the CTO of 'Snakecup' team continuously, and joined Proof-Of-Concept (POC) Funding Scheme 2022 in PolyU .
- Admitted to POC pith competition and won Dr. Winnie S M Tang- PolyU Student Innovation & Entrepreneurship Scholarship.

**RS INNOVATORS CHAMPIONSHIP 2023** Hong Kong, China
*Team leader & AI model designer of team EA Dynamics* November 30, 2022 – April 05, 2023

- Lead team named EA Dynamics joined RS Innovators Championship 2023.
- Proposed a mode based on LSTM to provide new strategy of logistics arrangement based on logistics data, improving distribute center working efficiency.
- Lead the team become 10 finalist teams worldwide and joined the final round the worldwide competition.

## COMMUNITY SERVICES

**HABITAT GREEN IN EAST AFRICA** Rwamagana District, Rwanda
*Volunteer student helper and technician* July 22 – August 06, 2023

- Going to Rwanda to build small solar power grid for power supply for villages which didn't have electrical power.
- Working as a technician to build the grid infrastructure by assisting wiring, circuit connecting, and training local youth to install power generation system.
- Interviewed by Hong Kong Ta Kung Wen Wei Media Group, HKTKWW. Relevant article was published on HKTWW and Rwanda local media, IGIHE.

## SELECTED AWARDS

**DR. WINNIE S M TANG- POLYU STUDENT INNOVATION & ENTREPRENEURSHIP SCHOLARSHIP**

- 'Dr Winnie S M Tang-PolyU Student Innovation & Entrepreneurship Scholarship' is the funding support of PolyU Student Entrepreneurial Proof-of-Concept (POC) Funding Scheme since 2022. (POC) Funding Scheme is a funding initiative launched in 2017 as one of the key entrepreneurship education components at PolyU. POC Fund aims to instil a problem-driven innovation mindset into young talents through experiential learning and student-initiated research and development. This funding is awarded to the undergraduate or graduate team who built entrepreneurship project has significant potential of making impacts on Hong Kong society. Up to $20,000 for each awarded team, and we received &15,000. My project 'Snakecup' and her team is awarded for the potential great healthcare and sustainable development impacts on Hong Kong community from the project.

**PROFESSOR LEUNG TIN-PUI MEMORIAL SCHOLARSHIP**

- The selection is based on the overall merits in the fields of academic performance, extra-curricular activities as well as community services. This scholarship with $20,000 would be awarded to students with high GPA performance (3.5+),

Outstanding in extra-curricular activities and community services, and strong aspirations on engineering.

**THE HONG KONG POLYTECHNIC UNIVERSITY SCHOLARSHIP 2023/24**

- The selection is based on the overall merits in the fields of academic performance, extra-curricular activities as well as community services. This scholarship with $40,000 would be awarded to students with high yearly GPA performance (3.7+).